Government of Tamilnadu

Department of Employment and Training

Course : TNPSC Group I Mains Material

Subject : Role and impact of Science and Technology in the development of India

Topic : Advancements in the fields of Space

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ADVANCEMENT IN THE FIELD OF SPACE

Indian space programme encompasses research in areas like astronomy, astrophysics, planetary and Earth sciences, atmospheric sciences and theoretical physics. Achievements here include ASTROSAT (India's first dedicated Indian astronomy mission), Mars Orbiter Mission (India's first Interplanetary Mission), Chandrayaan-1 (India's first mission to the moon). Chandrayaan-2 and Aditya 1 are in the works.

ASTROSAT

ASTROSAT launched on 28 September 2015, is the first dedicated astronomical mission to study the outer space objects in X-ray (0.3 keV to 100keV), limited optical and UV (near and far) spectrum. The unique feature of the satellite is that it enables the observation of various astronomical bodies with a single satellite simultaneously.

Objectives:

1. Estimation of the magnetic field of neutron stars.
2. Study of binary star system having black holes and neutron star and understanding high energy processes related to them.
3. To study the high energy processes related to star birth regions lying beyond our galaxy.
4. Deep field survey of the universe on a limited scale in Ultraviolet spectrum.
5. To detect new sources of X-rays in the sky.
6. The data collected is telemetered to the ground station for processing. All the payloads are currently in working condition.

Mangalyaan – Mars Orbiter Mission

The Mars Orbiter Mission (MOM), also called *Mangalyaan* is a spacecraft orbiting Mars since 24 September 2014. It was launched on 5 November 2013 using PSLV-C25 by ISRO. It is India's first interplanetary mission, and ISRO has become the fourth space agency to reach Mars, after the Soviet Space program, NASA and the European Space Agency. It is also the first nation to reach Mars orbit on its first attempt, and the

first Asian nation to do so. The mission is a "Technology Demonstrator" project to develop the technologies for the design, planning, management, and operations of an interplanetary mission. It carries five instruments that will help advance knowledge about Mars to achieve its secondary, scientific objective. The spacecraft is currently being monitored from the Spacecraft Control Centre at ISRO Telemetry, Tracking and Command Network (ISTRAC) in Bangalore with support from Indian Deep Space Network (IDSN) antennae at Byalalu.

Objectives of Mangalyaan

Payload

The scientific payload consists of five instruments:

Atmospheric studies:

1. **Lyman-Alpha Photometer (LAP) –** a photometer that measures the relative abundance of deuterium and hydrogen from Lyman-alpha emissions in the upper atmosphere. Measuring the deuterium/hydrogen ratio will allow an estimation of the amount of water loos to outer space.
2. **Methane Sensor for Mars (MSM)** – will measure methane in the atmosphere of Mars, if any, and map its sources.

Particle environment studies:

3. **Mars Exospheric Neutral Composition Analyser (MENCA) –** is a quadrupole mass analyser capable of analysing the neutral composition of particles in the exosphere.

Surface imaging studies:

4. **Thermal Infrared Imaging Spectrometer (TIS) –** will measure the temperature and emissivity of the Martian surface, allowing for the mapping of surface composition and mineralogy of Mars.
5. **Mars Colour Camera (MCC) –** will provide images in the visual spectrum, providing context for the other instruments.

ISRO plans to send a follow-up mission with a greater scientific payload to Mars between 2018 and 2020. This mission will likely consist of a lander and rover, rather than being orbiter only.

Chandrayaan-I (Mission to Moon)

Chandrayaan-1 was India's first unmanned lunar probe. It was launched by the ISRO in October 2008 and operated until Aug 2009. The mission included a lunar orbiter and an impactor. The vehicle was successfully inserted into lunar orbit on 8 Nov. 2008.

On 14 Nov. 2008, the Moon Impact Probe separated from the Chandrayaan orbiter and struck the South Pole in a controlled manner, making India the fourth country to place its flag on the Moon after the Soviet Union, the United States and Japan. The probe impacted near Shackleton Crater ejecting underground soil that could be analyzed for the presence of lunar water ice. It carried high-resolution remote sensing equipment for visible, near infrared, and soft and hard X-ray frequencies. Over a two-year period, it surveyed the lunar surface to produce a complete map of its chemical characteristics and three-dimensional topography.

The Polar Regions are of special interest as they might contain ice. Among its many achievements was the discovery of the widespread presence of water molecules in lunar soil. Although the mission was less than ten months in duration, and less than half the intended two years in length, a review by scientists termed the mission successful, as -it had completed 95% of all primary objectives, consisting of:

1. To construct the complex spacecraft with 11 scientific instruments.
2. To place the spacecraft in a circular orbit around the Moon by orbit rising manoeuvres from a near earth orbit.
3. To place the Flag of India on the Moon.
4. To carry out imaging operations and to collect data on the mineral content of the lunar soil.
5. To set up a deep space tracking network and implement the operational procedures for travel into deep space.

Payloads: The scientific payload had a total mass of 90 kg and contained five Indian instruments are six foreign instruments.

- **Terrain Mapping Camera (TMC)** is a CCD camera which was used to produce a high-resolution map of the Moon. The aim of this instrument was to-completely map the topography of the Moon When used in conjunction with data from Lunar Laser Ranging Instrument. (LLRI), it can help in better understanding of the lunar gravitational field as well.
- **Hyper Spectral Imager (HySI)** performed mineralogical mapping in the 400-900 nm band.
- **Lunar Laser Ranging Instrument (LLRI)** determines the height of the surface topography by sending pulses of infrared laser light towards the lunar surface and detecting the reflected portion of that light.
- **Moon Impact Probe (MIP)** developed by ISRO, is an impact probe which consisted of a C-band Radar altimeter for measurement of altitude of the probe, a video imaging system, for acquiring images of the lunar surface and a mass spectrometer for measuring the constituents of the lunar atmosphere.

The American Institute of Aeronautics and Astronautics (AIAA) has selected ISRO's Chandrayaan-1 mission as one of the recipients of its annual AIAA SPACE 2009 awards, which recognizes key contributions to space science and technology. The International Lunar Exploration Working Group (ILEWG) chose the Chandrayaan-1 team for giving the International Cooperation award.

Chandrayaan-2

Chandrayaan-2 is India's second lunar exploration mission. Developed by the ISRO, the mission is planned to be launched to the Moon by a Geosynchronous Satellite Launch Vehicle (GSLV), includes a lunar orbiter, a lander and a lunar rover, all developed by India. India is gearing up to launch Chandrayaan-2, in January 2018, India decided to go all alone in its second lunar missions, Chandrayaan-2, which was originally proposed as an Indo-Russia venture, as Russia pulled out after one of its mission failed. According to ISRO, this mission will use and test various new technologies are conduct 'new' experiments. The wheeled rover will move on the lunar

surface and will pick up soil or rock samples for on-site chemical analysis. The data will be sent to Earth through the Chandrayaan-2 orbiter.

Space Craft

The mission was originally planned to fly on a GSLV Mk II with an approximate lift-off weight of 2650 kg. But due to design modifications the weight increased to 3290 kg. Now, it is planned to be launched in a window of January to March 2019, from Satish Dhawan Space Centre, Sriharikota Using GSLV-MK-III launch vehicle which has a payload capacity of 4 tons.

The main payloads are the following:

Orbiter Payloads

1. **Large Area Soft X-ray Spectrometer (CLASS)** and **Solar X-ray monitor (XSM)** for mapping major elements present on the lunar surface.
2. **L and S – band Synthetic Aperture Radar (SAR)** for probing the first few tens of metres of the lunar surface for the presence of different constituents, including water ice. SAR is expected to provide further evidence confirming the presence of water ice below the shadowed regions of the Moon.
3. **Imaging IR Spectrometer (IIRS)** for mapping of lunar surface over a wide wavelength range for the study of minerals, water molecules and hydroxyl present.
4. **Neutral Mass Spectrometer (ChACE-2)** to carry out a detailed study of the lunar exosphere.
5. **Terrain Mapping Camera-2 (TMC-2)** for preparing a three-dimensional map essential for studying the lunar mineralogy and geology.

Lander: Unlike Chandrayaan-1's lunar probe, which impacted the Moon's surface, the lander will make a soft landing.

Rover: The wheeled rover will move on the lunar surface and will pick up soil or rock samples, perform chemical analysis and send data to Earth through the Chandrayaan-2 orbiter. The rover will weight 30 to 100 kg and will operate on solar power.

Rover Payloads

1. Laser-induced Breakdown, Spectroscope (LIBS)
2. Alpha Particle Induced X-ray Spectroscope (APIXS)

There are plans to include a seismometer to study Moon-quakes

Aditya-L1 (Spacecraft)

The original Aditya-L1 mission was conceived as a small satellite in a ~800km orbit carrying one payload to study solar corona. The mission has been enhanced and termed as Aditya-L1 mission carrying seven payloads, which was approved in 2015. It is scheduled for launch during 2020 timeframe by PSLV-XL Aditya-L1 can provide observations on the corona and also can provide observations on the solar chromosphere using a UV payload and, on the photosphere, and flares using X-ray payloads. These payloads taken together are expected to provide a comprehensive understanding of how solar flares originate and propagate. Also, the charged particle detectors and the magnetometer payloads can provide information on charged particles and the magnetic field which emanate from the eruptive events. To enable this, the Aditya-L1 spacecraft is to be placed in a halo orbit around the **Sun-Earth Lagrangian point 1 (L1)** which is about 1.5 million km from the Earth.

Lagrange point

Lagrange point is a position in space where combined gravitational forces of two large bodies, such as Earth and Sun of Earth and Moon, equal centrifugal force felt by a much smaller third body. The interaction of these forces creates a point of equilibrium where spacecraft may be "parked" to make an observation. These points are named after Joseph-Louis Lagrange, an 18th century mathematician. There are five such points between Earth and Sun namely – L1, L2, L3, L4, L5 in these orbits, the satellite requires very little energy to maintain its orbit, and it will not be eclipsed from the Sun.

Indian Human Spaceflight Program

The Indian human spaceflight program is a proposal by the ISRO to develop and launch the ISRO Orbital Vehicle, which is carry a two-member crew to Low Earth Orbit. Recent reports indicate that human spaceflight will occur after 2019, on a GSLV-

Mk ILL. Since the ISRO does not have a human-rated launch vehicle. ISRO will set up a full-fledged training facility in Bangalore for training astronauts. ISRO is planning to build a third launch pad at Sriharikota for manned missions with extra facilities like entry into the crew capsule and an escape chute. The trials for the manned space missions began with the 600 kg Space Capsule Recovery Experiment (SRE), launched using the Polar Satellite Launch Vehicle (PSLV) rocket, and safely returned to Earth 12 days later. This demonstrates India's capability to develop heat-resistant materials necessary for re-entry technology.

Description and Development

The major objective of manned mission program is to develop the fully autonomous three-ton ISRO Orbital Vehicle spaceship to carry a 2-member crew to orbit and safe return to the Earth after a mission duration of few orbits. The extendable version of the spaceship will allow fights up to seven days, rendezvous and docking capability with space stations or orbital platform. About 16 minutes after lift-off from the SDSC, Sriharikota, the rocket will inject the Orbital Vehicle into an orbit, 300 km-40 km from the Earth. The capsule would return for a splashdown in the Bay of Bengal.

The training facility at Bangalore will prepare personnel both for first orbital flights on board on Orbital Vehicle (OV) and for future manned Moon missions which will land Indians on the Earth's natural satellite after 202. This centre will train the selected astronauts in in rescue and recovery operations, surviving in a zero-gravity situation, the study of the radiation environment and for the long journey in the space through water simulation.

India would receive assistance in crew-selection and training from Russia under an agreement signed between the two countries. In March 2008. One option which was studied was light of an Indian astronaut aboard a Soyuz capsule in preparation for the Indian mission. However, in October 2010 this option was given up.

ISRO will build centrifuges to train astronauts on the high-gravity acceleration which occur when the vehicle lifts off. It also plans to build a new launch pad at a cost of `600 crore (US$132 million). It would be the third launch pad at the Satish Dhawan Space Centre on the east-coast in Andhra Pradesh.

Progress Related to Human Space Programme

Development of the launch vehicle capable of carrying crew to space, crew module, a mechanism to safely withdraw the crew in case of launch abort and training of the crew module is crucial for any manned mission in the space. In has successfully completed 2 developmental launches of the GSLV MK-III with 4-ton payload capacity. Now the GSLV MK-III is expected to be operational by mid-2019.

In December 2014 ISRO successfully tested the crew module also known as CARE (Crew/Module Atmospheric Re-entry Experiment). For the third requirement of safe evacuation of crew module in any exigency, ISRO conducted Pad Abort Test on 5 July 2018. The details are discussed below.

Care Module

Crew Module (CM) is identified as the payload in GSLV MK-III X/CARE Mission launched on 18 December 2014. CARE is the acronym for Crew module Atmospheric Re-entry Experiment. The mission would be used as a platform for testing the re-entry technologies envisaged for Crew Module including validating the performance of parachute-based deceleration system. CARE is expected to enhance the understanding of blunt body re-entry aero-thermodynamics and parachute deployment in a cluster configuration.

The external configuration of the Crew Module to be flight tested in GSLV MK-III is same as that of manned flight. The Crew Module (CM) is separated from the Launch Vehicle at an altitude of 126 Km, re-entered Earth's atmosphere at about 80 km and descended further in ballistic mode. Beyond 80 km, CM followed an uncontrolled re-entry trajectory and impacted at sea about 180 km from Andaman and Nicobar Island from where it was recovered by Indian Coast guard.

Pad Abort Test

ISRO carried out a major technology demonstration on July 05, 2018, the first in a series of tests to qualify a Crew Escape System, which is a critical technology relevant for human spaceflight. The Crew Escape System is an emergency escape measure.

Designed to quickly pull the crew module along with the astronauts to a safe distance from the launch vehicle in the event of a launch abort. The first test (Pad Abort Test) demonstrated the safe recovery of the crew module in case of any exigency at the launch pad. The Crew Escape System along with the simulated crew module with a mass of 12.6 tonnes, lifted off at 07.00 AM (IST) at the opening of the launch window from its pad at Satish Dhawan Space Centre, Sriharikota. The test was over in 259 seconds, during which the Crew Escape System along with crew module soared skyward, then arced out over the Bay of Bengal and floated back to Earth under its parachutes about 2.9 km from Sriharikota. The crew module reached an altitude of nearly 2.7 km under the power of its seven specifically designed quick acting solid motors to take away the crew module to safe distance without exceeding the safe g-levels. The Crew escape system uses solid propellant based jettisoning motors to take the crew module away from the launch vehicle.

SCRAMJET ENGINE

A Scramjet is a supersonic Ramjet engine. Ramjet engine is an air-breathing engine that uses the atmospheric oxygen for the oxidation or burning of the fuel to generate power. The Scramjet engine also uses the atmospheric oxygen for this purpose, but it can use it at supersonic speed. While the normal rockets carry both fuel and oxidizer to generate the thrust required, the Scramjet engine needs to carry the fuel alone as it can use the atmosphere oxygen as the oxidizer. It compresses the atmospheric air due to its supersonic forward motion and use it to burn the fuel, thus reducing the weight of the rocket and increasing the payload capacity of the rocket.

But the challenge with the Scramjet engine is that it operates at supersonic speed. At such a high speed the flow of the air is very turbulent, and the control of the air flow and sustaining the combustion require a very high level of expertise in the field. The challenge is also that the Scramjet engine cannot start operation itself. It is required to be taken to the supersonic speed using another rocket. Once it reaches the supersonic speed, it starts operational.

Another limitation of the Scramjet engine is that it can only in the atmospheric zone of the rocket flight as there is no oxygen for the engine to operate beyond the atmosphere.

In 2016 ISRO successfully tested the experiment flight of the Scramjet engine which was attached to a Rohini sounding rocket. If successfully operationalized it can reduce the cost of the launch vehicles by more than half by eliminating the cost of oxidizer required and increasing the payload capacity by reducing the weight of launcher. It can revolutionize space tourism.

Indian Space Vision 2025:

The space vision 2025, a future roadmap for ISRO was unveiled at the Indian Science Congress Session in 2003, in Bangalore. The vision document spells out the steps to take our space programme to greater heights. As India has already achieved self-sufficiency in the fabrication of satellites, thus now the emphasis is on achieving self-reliance in launching capabilities.

It includes:

1. Satellite-based communication and navigation systems for rural connectivity, security needs and mobile services;
2. Enhanced imaging capability for natural resource management, weather and climate change studies;
3. Space science missions for a better understanding of solar system and universe;
4. Planetary exploration;
5. Development of Heavy lift launcher;
6. Reusable Launch Vehicles – Technology demonstrator missions leading to Two Stage to Orbit (TSTO); and
7. Human Space Flight

Questions:

1. Write a note on Mangalyaan and its Objectives.
2. Give a brief answer about Chandrayaan Missions.
3. Discuss about the India's human spaceflight program.